SiamMDM: An Adaptive Fusion Network With Dynamic Template for Real-Time Satellite Video Single Object Tracking

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***Abstract*— Tracking moving targets in satellite videos has attracted wide attention recently. However, the development of target tracking in satellite videos is much slower than that in general videos for the following key reasons. First, typical moving objects in satellite videos consist of few pixels and lose most of their appearance features, making it difficult for the tracker to distinguish the target from the background. Second, the appearance of satellite video objects often changes due to occlusion, illumination variation, or other factors. Classic Siamese tracking networks only use the first frame as the target template, leading to poor tracking results. Third, when the target is fully occluded, it is difficult for the tracker to recapture the target. To address the above problems, we propose a Siamese tracking network based on multiple (M) response map fusion and spatiotemporal constraints in this article. By generating response maps at different layers of the tracking network and fusing them adaptively, small objects in the satellite videos can be tracked more accurately. Furthermore, a dynamic (D) template update strategy is proposed to cope with possible changes in the appearance of objects in satellite videos, preventing the high dependence on the initial frame. To recapture the target, a score-guided target motion (M) trajectory prediction model is proposed. We call the proposed Siamese tracking network SiamMDM for short. We conducted complete experiments on SatSOT and SV248S, two large satellite video target tracking datasets. The results show that our method achieves state-of-the-art tracking performance while running at over 110 frames/s (FPS).**

***Index Terms*— Dynamic template (DT), map fusion, motion estimation (ME), object tracking, satellite video, Siamese network.**

I. Introduction

VISUAL object tracking is one of the most critical and challenging tasks in computer vision. In recent years, remote sensing satellite video technology has achieved tremendous development. At the end of 2013, the American company skybox launched the world's first video satellite SkySat-1, which can capture panchromatic video with a ground sample distance (GSD) of 1.1 m and a frame rate of 30 frames/s (FPS). In 2014, UrtheCast sent a high-resolution iris camera named Iris to the International Space Station and successfully installed it, which can shoot 4k ultrahigh-definition (UHD) full-color video on Earth with a resolution of 1 m. From 2015 to the present, China Changguang Satellite Corporation has successively launched a number of satellites to form the Jilin-1 satellite constellation. Currently, Jilin-1 satellite constellation has the ability to shoot 30 FPS RGB satellite videos with a GSD of 0.92 m and the coverage of each frame reaching 11.0 × 4.6 km [1].

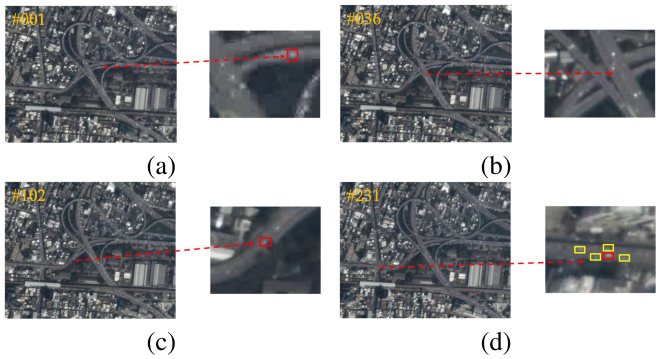


Fig. 1. Challenges in satellite video target tracking. (a) Vehicle in satellite video target tracking, occupying only 7 × 8 pixels. (b) Target vehicle is completely occluded by the bridge road. (c) Target vehicle is partially occluded by the shadow of the building. (d) Target vehicle is surrounded by similar-looking vehicles.

Because of the rapid development of remote sensing satellite technology [2], satellite video tracking is widely used in many fields, such as moving object analysis [3], traffic condition monitoring [4], visual surveillance [5], tiny object detection [6], fighting wildfires [7], and so on. Given the size and position of a target in the initial frame of a video sequence, the goal of tracking is to estimate the size and position of the target in subsequent frames [8]. In the practical application of target tracking, in addition to the tracking precision, the speed of the tracker is also an important indicator to measure the performance of trackers [9].

Compared with general video tracking, satellite video tracking faces particular challenges in the following aspects.

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1) *Smaller Target Size:* Compared with general video targets, targets in satellite videos are smaller. Some typical moving targets in satellite videos, such as vehicles, ships, and some aircraft, occupy only a few pixels in the picture. Also, the object only takes up about 0.01% of the entire satellite imagery [10]. As illustrated in Fig. 1(a), a vehicle to be tracked moving on the road occupies only 7×8 pixels, while the whole satellite image size is 1020×812 pixels. Since the target is very small, the trackers cannot obtain enough feature representation from the image. Therefore, some trackers that perform well in general video tracking cannot achieve satisfactory results in satellite video tracking.
2) *More Occlusion Issues:* When a moving target passes through overpasses, buildings, trees, and so on, the target will be partially or completely occluded. In the satellite video, as illustrated in Fig. 1(b) and (c), occlusion can easily cause tracking drift or interrupt the tracking. For some trackers, it is challenging to retrack the original target after losing it.
3) *More Severe Background Noise:* Moving targets are usually surrounded by many similar objects. These objects bring great difficulty for trackers to track the target continuously. As illustrated in Fig. 1(d), when the vehicle to be tracked moves to a traffic-congested area, some vehicles with similar appearance to the target vehicle appear around. It is difficult to distinguish them simply by the target appearance in a single frame image. Worse yet, in some cases, the color characteristics of the moving target and the background are similar (for example, a black vehicle driving on a black road), and the target will visually "disappear" in the background. It is difficult to distinguish the target from the background.

Our research hopes to improve the performance of satellite video target tracking by taking advantage of neural networks' ability to extract features and fully exploiting the movement of moving targets in satellite videos. In this article, we propose a Siamese network SiamMDM based on multiple response map fusion (M; MRMF), dynamic template (DT) update strategy (D), and motion trajectory prediction model (M). The main contributions of our work are summarized as follows.

1) We propose an MRMF method, which combines shallow response maps and deep response maps to solve the problem of inaccurate tracking caused by tiny targets in satellite video target tracking.
2) We propose a DT update module to cope with object appearance change problem, preventing the high dependence on the initial frame.
3) Considering that the motion of the satellite video target in a short time is almost linear and uniform, we propose a score-guided motion trajectory prediction model to solve the problem that the target cannot be continuously tracked when the target is completely occluded.
4) Extensive experiments have shown that our algorithm can track targets with the optimal precision score and success rate on both the SatSOT and SV248S datasets, reaching the state-of-the-art level while tracking at speeds far exceeding the requirement for real-time tracking.

The rest of this article is organized as follows. In Section II, we briefly introduce the related work on object tracking. Then, in Section III, we introduce the proposed object tracking network in detail. After that, we present results and analysis of the experiments in Section IV. Finally, we end this article with the conclusion in Section V.

II. Related Work

A. Correlation Filter-Based Methods

The correlation filter originally comes from the field of signal processing in the Fourier domain [11], and the basic idea is to design a filter template and use it to do the correlation operation with the target candidate region, and the position of the maximum output response is the target position of the current frame. Minimum output sum of squared error (MOSSE) [12] first introduced the correlation filtering method to the field of target tracking, which uses multiple samples of the target as training samples to generate better filters. MOSSE takes the minimization of the sum of squares error as the objective function and uses m samples to find the least squares solution. To address the problem of sample redundancy caused by sparse sampling in the MOSSE algorithm, exploiting the circulant structure of tracking-by-detection with kernels (CSK) [13] extends the ridge regression, the approximate dense sampling method based on circular shifting, and the kernel method. MOSSE and CSK process single-channel grayscale images and introduce circular shifts and fast Fourier transforms, which greatly improve the computational efficiency of the algorithm. Color name (CN) [14] extends the multichannel color to optimize the CSK method. KCF [15] also expands the number of negative samples using a circular matrix construction method based on CSK and enhances the performance of the tracker in this way. Meanwhile, KCF expands the multichannel features on the basis of CSK and uses histogram of oriented gradient (HOG) multichannel features to replace grayscale features. DSST [16] introduces the target scale estimation into the CSK algorithm. The algorithm learns discriminative correlation filtering (DCF) based on the scale pyramid description and learns two different filters, the scale estimation filter and the position filter, respectively. To further improve the representation of image features, the C-COT [17] algorithm introduces depth features into the correlation filtering. ECO [18] improves on C-COT by accelerating C-COT in three aspects: model size, sample set size, and update strategy.

In addition to these correlation filter-based tracking methods based on conventional three-channel color images, several methods are exploring the potential of hyperspectral images (HSIs) in the tracking field. An HSI is an image that records both spatial and continuous spectral information of a target in a scene [19]. Because of the target material information contained in a large number of bands of HSIs, trackers based on HSIs work better in some difficult conditions, such as complex background scenes, rapid changes in the appearance of the target, and so on.

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Uzkent et al. [20] proposed a deep hyperspectral KCF-based tracker (DeepHKCF) to efficiently track objects using an adaptive multimodal hyperspectral sensor. Xiong et al. [21] designed a dynamic material-aware tracking (DMT) method and adopted abundances and spectral–spatial HOG (SSHOG) as material features for tracking. Xiong et al. [22] proposed a material-based hyperspectral tracker (MHT), which embeds spectral–spatial information of HSIs into the histogram of multidimensional oriented gradients and then combines it with the global material abundance features. Chen et al. [23] put forward a correlation filter object tracker based on fast spatial–spectral features (FSSF) to realize robust, real-time object tracking in hyperspectral surveillance video. Chen and Zhao [24] constructed a novel mosaic SSHOG (MSSHOG) descriptor and embedded it in correlation filters to make full use of the information in HSIs for tracking.

B. Deep Learning-Based Methods

Deep learning has demonstrated extraordinary performance in several vision domains and, thus, has attracted enough attention in visual object tracking. SINT [25] pioneered the first conversion of the target tracking problem to a patch block matching problem. SiamFC [26] is an end-to-end Siamese network that uses the translational invariance of full convolution for matching between the template region and the search region. It can run faster at 86 FPS at three scales and get better results at five scales. CFNet [27] combines correlation filtering and convolutional neural networks and enables end-to-end training of correlation filtering. SA-Siam [28] requires the two branch networks of the Siamese network to be trained separately to learn different features while adding an attention mechanism and fusion of multilayer features in the template branch. RASNet [29] uses multiple attention mechanisms to weigh the space and channel of SiamFC features to decompose the coupling of feature extraction and discriminative analysis, which is used to improve the discriminative power of the network. SiamRPN [9] applies the region proposal network (RPN) [30] module from target detection to the field of target tracking, converting the original similarity calculation problem into a regression and classification problem. The network improves accuracy while increasing speed and begins to experiment with larger datasets in the target tracking network training process. DaSiamRPN [31] pays more attention to the possible effects of the input sample data on the network performance and mainly deals with the positive and negative sample block imbalance problem and the richness of the sample blocks. SiamMask [32] adds the mask branch to SiamRPN, enabling both video target tracking and video target segmentation. SiamDW [33] analyzed the impact of each part of the Siamese target tracking network design on the results and designed the cropping-inside residual units (CIR) module to make the network wider and deeper based on the analysis. SiamRPN++ [34] introduces hierarchical aggregation to exploit multilevel features and depthwise cross correlation to reduce computational costs and memory. As Transformer (TF) [35] continues to grow in the computer vision field, more and more work is being done to introduce TF into the field of target tracking. Inspired by TF, TransT [36] proposes an attention-based feature fusion network that uses Siamese networks to extract features and uses attention to combine template region features and search region features efficiently. Stark [37] proposes a new TF architecture dedicated to visual tracking. It can capture the global feature dependencies of spatial and temporal information in video sequences. The whole approach does not require postprocessing steps, such as cosine windows, bounding box smoothing, and so on, which greatly simplifies the existing tracking pipeline. HiFT [38] proposes a new hierarchical feature converter to learn the relationship between multilevel features. TCTrack [39] is the first attempt to introduce temporal information in the feature extraction dimension and integrate temporal information in the similarity graph dimension based on TF.

Deep learning-based tracking methods require a large amount of high-quality labeled data to drive the training and evaluation of trackers. To facilitate the further development of deep learning in the field of tracking, more and more annotated tracking datasets for different tracking tasks and different tracking scenarios are proposed. In the field of single object tracking (SOT), Muller et al. [40] proposed the TrackingNet dataset, the first large-scale dataset and benchmark for object tracking in the wild. Fan et al. [41] presented the LaSOT dataset, a high-quality benchmark for large-scale SOT consisting of 1400 sequences with more than 3.5M frames. Huang et al. [42] released the large high-diversity database GOT10k, which aimed to provide a unified training and evaluation platform for developing class-agnostic, generic purposed short-term trackers. Various tracking datasets are also proposed in the field of multiple object tracking (MOT). Dave et al. [43] introduced a similarly diverse dataset for tracking any object (TAO). Voigtlaender et al. [44] provided two new datasets with temporally consistent object instance segmentations based on the KITTI and MOTChallenge datasets for training and evaluating methods that tackle the multiobject tracking and segmentation (MOTS) task. Cui et al. [45] introduced the DG-Labeler and DGL-MOTS datasets to facilitate the training data annotation for the MOTS task. Wen et al. [46] presented a large-scale multiobject tracking benchmark (UA-DETRAC) consisting of 100 videos with rich annotations. Yu et al. [47] presented an unmanned aerial vehicle (UAV) dataset with 100 videos captured in complex scenarios to facilitate research in the UAV field.

Deep learning-based approaches with different implementation strategies also flourish in hyperspectral target tracking. Liu et al. [48] proposed a dual deep Siamese network framework that designed a hyperspectral target-aware module to mine the spectral information during the target prediction and introduced a spatial–spectral cross-attention module to further fuse the deep spatial and spectral features. Zhao et al. [49] designed a TF-based fusion tracking network (TFTN) for hyperspectral and RGB fusion tracking to improve the tracking performance by aware-aggregating the information of different modalities through the modality reliability. Li et al. [50] introduced an autoencoder-like band attention block to learn the dependencies among bands and generate bandwise weights, then divided HSIs into a number of three-channel images according to the band weights, and fed them into a deep color tracking network,

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producing several weak trackers. Liu et al. [51] proposed an anchor-free Siamese network for hyperspectral video target tracking (HA-Net). A spectral classification branch is introduced to the anchor-free Siamese network to increase the network's ability to identify objects. Lei et al. [52] proposed a spatial–spectral cross-correlation embedded dual-transfer network (SSDT-Net), which uses transfer learning to transfer the knowledge of traditional color videos to the HS tracking task to address the problem of limited annotation samples.

C. Visual Object Tracking in Satellite Videos

Since there are considerable differences between satellite video target tracking and general optical target tracking, many researchers are investigating tracking methods specifically for satellite video targets.

Some algorithms have proposed solutions for target motion estimation (ME), mainly for satellite videos where the background is relatively stationary and the moving target shows linear motion in a short time. Xuan et al. [53] proposed a new ME algorithm by combining the Kalman filter and motion trajectory averaging, and they use this ME strategy to mitigate the boundary effect of KCF. Zhu et al. [11] proposed a deep Siamese network incorporating an interframe difference centroid inertia motion (ID-CIM) model and using the ID-CIM model to alleviate model drift. Shao et al. [3] proposed a velocity correlation filter, employing velocity feature and inertia mechanism to construct a kernel correlation filter for satellite video target tracking. Zhang et al. [54] proposed a multifeature correlation filter with ME to deal with occlusion, low resolution, and insufficient information. Guo et al. [55] proposed a high-speed correlation filter-based tracker for object tracking on satellite videos. It takes advantage of the global motion characteristics of the moving target in satellite videos to constrain the tracking process.

Some algorithms try to use frame difference or optical flow methods to enhance the precision of satellite video target localization. Du et al. [10] proposed a multiframe optical flow tracker for object tracking in satellite videos. They fused the Lucas–Kanade optical flow method with the HSV color system and integral image to track the targets in the satellite videos. Experiments show that a better interpretation can be obtained by using the multiframe difference method in the optical flow tracker.

Some algorithms focus on solving the problems caused by target rotation (ROT) in satellite videos. Chen et al. [56] proposed a novel ROT adaptive tracker with motion constraints to boost satellite video object tracking from ROT and translation. Xuan et al. [57] proposed a rotation-adaptive correlation filter (RACF) tracking algorithm.

Some algorithms try to solve the problem of target tracking in satellite videos from other perspectives, such as super-resolution, background processing, and so on. Shao et al. [58] designed a lightweight parallel network HRSiam with a high spatial resolution to locate the small objects in satellite videos. Wang et al. [59] enhanced target tracking for satellite video technology by using sample training strategy and sample characterization. Zhang et al. [60] proposed to learn the motion and background of the target in order to help the trackers recognize the target with higher precision.

III. Proposed Approach

A. Overall Architecture

Fig. 2 shows the overall framework of our proposed SiamMDM method, which consists of three modules: the MRMF module, the DT update module, and the motion trajectory estimation module.

First, the template branch of SiamMDM takes the target region of the first frame as input. The search branch takes the search region of the current frame as input. These two branches share the same feature extraction network. Then, the MRMF module performs cross-correlation operations on different layers of the Siamese network to get multiple response maps. It adaptively fuses the response maps to obtain the Fused Map I. In order to reduce the absolute dependence on the first frame, a DT module is added to the network. The DT module decides whether the DT needs to be updated every K frames, and then performs cross-correlation operations between the DT and the search region to get Fused Map II after MRMF. The network combines Fused Map I and Fused Map II to get Fused Map III. After that, we feed the previously cached target position into the motion model (MM) and predict the expected value of the target position in the current frame based on the motion characteristics of moving targets in the satellite video. Finally, SiamMDM adaptively fuses the target position in Fuse Map III with the target position predicted by the MM to obtain the final target position.

The network uses the following equation as the loss function:

$$l(y, v) = \log(1 + \exp(-yv)) \tag{1}$$

$$L(y, v) = \frac{1}{|D|} \sum_{u \in D} l(y[u], v[u]) \tag{2}$$

where v is a single response value output by the network and y is the label of the corresponding position, $y \in \{-1, 1\}$. D is the generated response map, u is a value in D, and $|D|$ is the response map size. The labels of the response maps are labeled as follows:

$$y[u] = \begin{cases} +1, & \text{if } k\|u - c\| \le R \\ -1, & \text{otherwise.} \end{cases} \tag{3}$$

Among them, c is the center of the response map, u is any point in the response map, and k is the zoom factor of the response map after passing through the network. The pseudocode of our SiamMDM algorithm can be seen in Algorithm 1.

B. MRMF Strategy

As mentioned earlier, the first problem faced by satellite video target tracking is that targets in satellite video occupy few pixels [61]. These pixels usually form a point or line shape, missing texture, and unique shape information. Therefore, the network is unlikely to extract enough features from the original image, making it difficult to obtain competitive tracking results using only the feature maps from the network's last layer. So, we consider using multilevel features of Siamese

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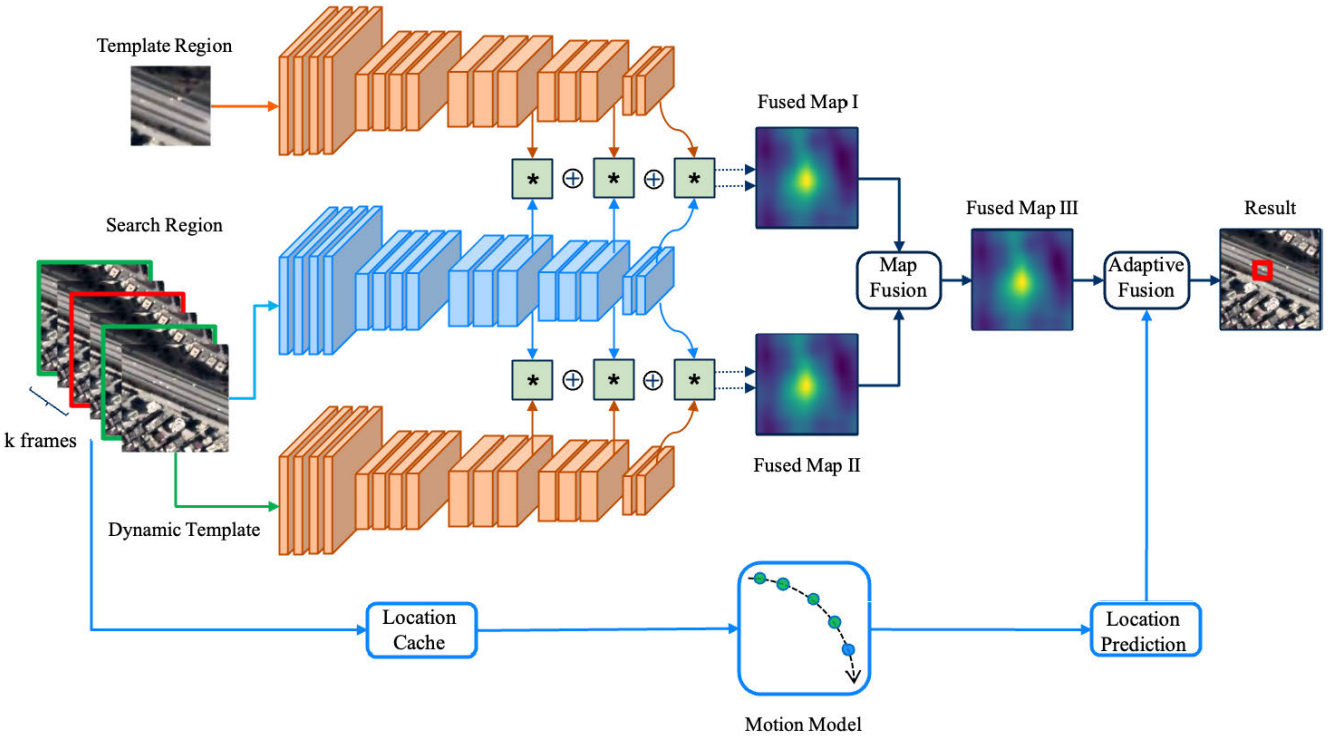


Fig. 2. Architecture of our proposed SiamMDM framework. First, we use an MRMF strategy in the last three layers of the neural network. Then, we obtained the fusion response maps from the initial template branch and the DT branch, respectively. Third, we fuse the above two response maps. After that, we use the MM to correct the final target position predicted by the tracker, and we get the final position of the target.

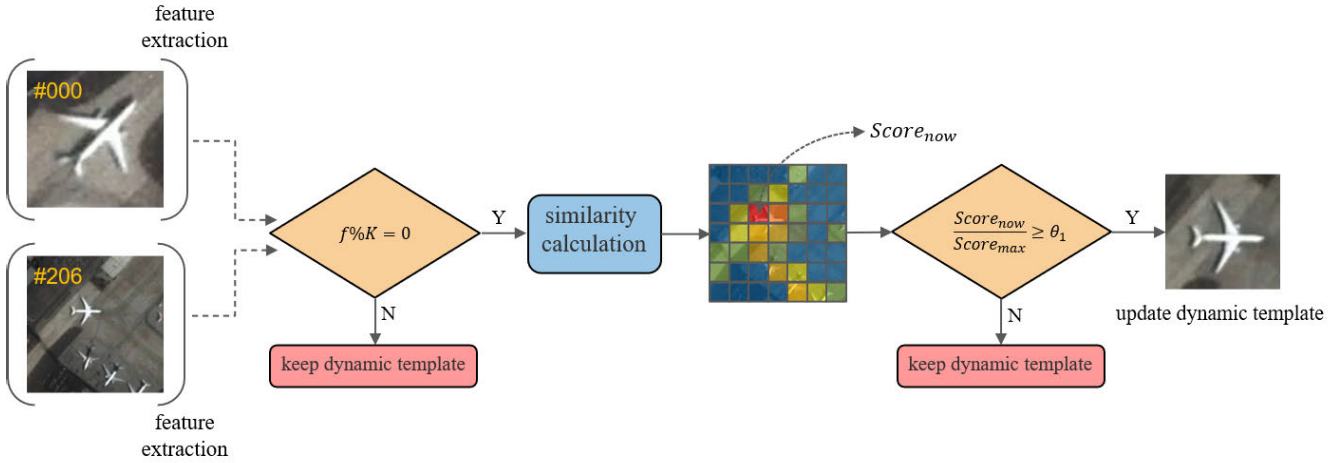


Fig. 3. Flowchart of updating the DT. $\text{Score}_{\text{now}}$ represents the highest value in current response map.

network for tracking. As pointed out in [7] and [62], the features in the earlier layers retain higher spatial resolution and more fine-grained spatial details, which is useful for precise localization, while features in the latter layers capture more semantic information, which is robust to ROT and appearance changes. Therefore, we propose an MRMF module. First, this module performs cross-correlation operations on the shallow, middle, and deep layers of the Siamese network to obtain multiple response maps. The specific similarity calculation is shown in the following equation:

$$g_l(Z_1, X) = f_l(Z_1) * f_l(X) + b \tag{4}$$

where Z_1 represents the template region in the initial frame, X represents the search region in the current frame, $*$ is the cross-correlation operation, and b denotes a bias signal, which adds $b' \in \mathbb{R}$ to each position of the response map.

After that, response maps obtained from different layers are fused using a weighted approach to obtain the Fused Map I. The fusion equation can be written as follows:

$$P_{\text{fusion}} = \sum_{l=3}^{5} \alpha_l g_l(Z_1, X) \tag{5}$$

where α_l is the weight corresponding to each response map. Instead of setting weight parameters manually, the above

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Algorithm 1 Procedure of the Proposed SiamMDM Method

Input:

P_0: The target position in the initial frame of the video;
f: Video frame index;
N: Trajectory length used by the Motion Model;
K: Frequency of updating dynamic templates;

Output:

P_n: The target position in the nth frame of the video;

for $f <= N$ **do**
  if $f == 1$ **then**
    Learn the template feature of the target location P_0
  else if $f \ \% \ K == 0$ **then**
    Update the dynamic template according to equation (7);
    Learn the dynamic template feature of the target;
    Calculate the final tracking result using Algorithm 2;
  else
    Calculate the final tracking result using Algorithm 2;
  end if
end for

Algorithm 2 Calculate the Final Tracking Result

Input:

P_0: The target position in the initial frame of the video;
f: Video frame index;
N: Trajectory length used by the Motion Model;
K: Frequency of updating dynamic templates;

Output:

P_n: The target position in the nth frame of the video;

Calculate the response maps between the search region of the current frame and the template region of the initial frame;
Calculate the fused response map I by Equation (5);
if $f > K$ **then**
  Calculate the response maps between the search region of the current frame and the template region of the dynamic template frame;
  Calculate the fused response map II by Equation (5);
  Calculate the locations $P_{\text{Map I}}$ and $P_{\text{Map II}}$ with the largest response values in response map I and response map II, respectively;
  if $\left\| P_{\text{Map I}} - P_{\text{Map II}} \right\| < \theta_2$ **then**
    Get the maximum response point location $P_{\text{Map III}}$ of the response map by Equation (8);
  else
    $P_{\text{Map III}} = P_{\text{Map I}}$
  end if
end if
if $f > N$ **then**
  Calculate the desired displacement of the target in the x, y direction with Equation (11) and Equation (12);
  Get the target expected position P_{MM} of the Motion Model's prediction;
  get the final tracking result P_{Final} with Equation (15) and Equation (16);
end if

three weight parameters are automatically determined by the network. We take the weight parameters before each response as part of the network, so that they are optimized together with the network during the training process.

C. DT Update Strategy

Satellite video objects are prone to significant changes in shape and brightness due to occlusion and motion blur. Although the initial frame of the video contains a lot of highly reliable information about the object's appearance, it is inappropriate for the network to use only the initial frame as a template frame during tracking. The reason is that when the target appearance changes significantly due to some factors, the tracking network can easily lose the target or the tracking drifts to similar surrounding objects. On the other hand, the previous frames of the current frame are very relevant to the current frame. They contain more helpful information about the current change in the target appearance. However, if the network uses the previous frame of the current frame of the video as the template frame, when the target track drifts in the previous frame, tracking drift occurs in all subsequent frames of the video [63]. In addition, feeding each frame as a DT into the feature extraction network would be time-consuming.

Therefore, we propose a target DT update module. The flowchart of updating the DT is shown in Fig. 3. The target DT update module decides whether to update the DT every K frames. It uses the dynamic and initial frame templates to adapt the target tracking requirements of the current frame and improve the robustness of the target tracking. The specific implementation process of the DT update strategy is as follows. After feeding each video frame into the Siamese network, the Siamese network will obtain the max similarity value between the current frame and the initial frame template, which is denoted as $\text{Score}_{\text{now}}$. The theoretical maximum value of $\text{Score}_{\text{now}}$ is denoted as $\text{Score}_{\text{max}}$, which is obtained by doing a similarity calculation between the initial template region and itself according to (4). We define the reliability coefficient of the Siamese network as follows:

$$\alpha = \frac{\text{Score}_{\text{now}}}{\text{Score}_{\text{max}}}. \tag{6}$$

A higher reliability coefficient indicates that the object identified by the Siamese network in the current frame is more likely to be the target. Whenever K frames pass, the DT update module decides whether to update the previous DT frame to the current frame. The equation for the decision rule is as follows:

$$\begin{cases} Z_D = T_{\text{now}}, & \text{if } \alpha > \theta_1 \\ Z_D = Z_{D-1}, & \text{if } \alpha \le \theta_1 \end{cases} \tag{7}$$

where Z_D represents the current DT frame, T_{now} represents the current frame of the video, and θ_1 represents a threshold. When the reliability coefficient is greater than the threshold, change the DT frame to the current frame. Otherwise, the DT is not updated, and the network still uses the previous DT frame. By updating the DT in this way, the DT not only contains

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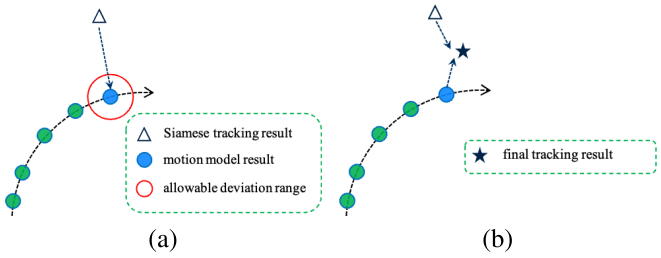


Fig. 4. (a) Mechanism of the traditional MM. (b) Mechanism of our MM.

much information in the initial frame of the template but also enriches some changes in the target's appearance over time. Combining the initial and dynamic frame templates enables better and more robust tracking.

After the MRMF operation, Fused Map I and Fused Map II are obtained from the initial frame template branch and the dynamic frame template branch, respectively. We fuse the above two response map using the following equation:

$$P_{\text{Map III}} = \eta P_{\text{Map I}} + (1-\eta) P_{\text{Map II}} \tag{8}$$

where η represents the weight coefficient of the two response maps.

Considering that when other objects have similar appearances around the target, the maximum position of the response map obtained from the initial template frame branch and the DT frame branch may correspond to different objects. In this case, simply fusing the two response maps does not give better results but leads to the "maximum diffusion" phenomenon, and the tracking results may be worse. Therefore, we specify the following equation:

$$\begin{cases} \eta = 1, & \text{if } D_{\text{drift}} > \theta_2 \\ \eta \neq 1, & \text{if } D_{\text{drift}} \leq \theta_2. \end{cases} \tag{9}$$

D_{drift} represents the Euclidean distance between the position P_1 corresponding to the maximum response value in Fused Map I and the position P_2 corresponding to the maximum response value in Fused Map II

$$D_{\text{drift}} = \| P_1 - P_2 \|. \tag{10}$$

That is, when the distance between the response maxima in Fused Map I and Fused Map II exceeds a certain threshold, we force the network to trust only the initial number of template frames, and vice versa, the network still operates according to (5).

D. Motion Model

Unlike targets and their motion in general optical target tracking, targets in satellite videos hardly undergo deformation [55], and the object motion in satellite videos is relatively simple and clear. Furthermore, because the time between every two frames of the satellite videos is short, it can be assumed that moving objects, such as vehicles, are moving in a uniform linear motion [53]. The traditional motion prediction model is shown in Fig. 4(a). It obtains the desired position of the target in the current frame by the positions of the target in the previous frames. Then, the model sets a range around the desired position. If the prediction result of the Siamese network is within this range, then the result remains unchanged. If the prediction result is outside the range, then the result of the MM is used as the final prediction result. But, such an approach has disadvantages. First, due to the camera shaking and the target changing direction in a short time, the target may leave the trajectory, and then, forcibly changing the prediction will degrade the performance instead. Second, the range parameter will affect the final result. If the deviation range is large, the effect of the MM will be weak; if the deviation range is small, the constraint of the MM will be very strong and results in unexpected shift in predicted position. Manually setting the range parameter is time-consuming.

Therefore, we propose a score-guided motion prediction model (hereinafter referred to as "MM") to solve the predicted position drift problem. The mechanism of our MM is shown in Fig. 4(b). First, we cache the target's location in the previous N frames predicted by the network and calculate the distance of the target moving in the x-direction and the y-direction between two adjacent frames, respectively. The specific operation rules of the MM are shown as follows:

$$\triangle x_t = \frac{1}{N} \sum_{i=1}^{N} \triangle x_{t-i} \tag{11}$$

$$\triangle y_t = \frac{1}{N} \sum_{i=1}^{N} \triangle y_{t-i} \tag{12}$$

where Δx_t represents the difference between the predicted target position and the current target position in the x-direction and Δy_t represents the difference between the predicted target position and the current target position in the y-direction. Then, the results of the MM for the Nth frame are shown in the following equation:

$$X_{\text{MM}} = X_{t-1} + \triangle x_t \tag{13}$$

$$Y_{\text{MM}} = Y_{t-1} + \triangle y_t \tag{14}$$

where X_{MM} and Y_{MM} represent the MM result of current frame and X_{t-1} and Y_{t-1} represent the final tracking result of previous frame.

Therefore, we can get the final tracking result as shown in the following equation:

$$X_t = \alpha X_{\text{Map III}} + (1-\alpha) X_{\text{MM}} \tag{15}$$

$$Y_t = \alpha Y_{\text{Map III}} + (1-\alpha) Y_{\text{MM}}. \tag{16}$$

The calculation equation for α is shown in (6), where $\text{Score}_{\text{now}}$ and $\text{Score}_{\text{max}}$ here represent the current response maximum and the maximum possible value of $\text{Score}_{\text{now}}$ in Fused Map III, respectively.

When the reliability coefficient α is sufficiently high, we consider the prediction results of the Siamese network to be reliable and, therefore, reduce the involvement of the MM in the final results. Conversely, if the reliability coefficient α is low, we consider that the tracking of the Siamese network encounters difficulties, and increase the involvement of the MM in the final prediction results. With this score-guided approach, the network can adaptively adjust the involvement of the MM in the final prediction results.

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Fig. 5. Four typical targets in the SatSOT and SV248S datasets.

IV. Evaluation

This section introduces the details of the experiments, including the datasets and the evaluation methods. We present the results of our SiamMDM method on the dataset, with comparisons to the state-of-the-art object tracking algorithms. Ablation studies are also provided to analyze the effects of various modules in the proposed network.

A. Experimental Setup

1) Dataset: In this study, the SatSOT [61] and the SV248S [64] satellite video datasets are used to comprehensively evaluate the effectiveness of the proposed method. Four typical targets in the above two datasets are shown in Fig. 5. There are cars, planes, ships, and trains. Among them, train is a unique category in the SatSOT dataset, and the SV248S dataset only contains the first three categories. The SatSOT dataset consists of 105 video sequences. The average video length of SatSOT is 263 frames. The shortest video consists of 120 frames, while the longest video contains 750 frames. The SatSOT dataset contains 26 trains, 65 cars, nine planes, and five boats, totaling 27 664 frames. In these video sequences, the average bounding box sizes occupied by trains, cars, planes, and ships are 39 566.3, 112.4, 1714.3, and 438.5, respectively. These video sequences include 11 different challenge attributes. The attributes and corresponding definitions are shown in Table I. Each video is annotated with the most affected attributes.

The SV248S dataset consists of 248 video sequences that range in length from 500 to 753 frames, and they last for 20–30.12 s with the frame rate at 25 FPS. This dataset focuses on the tracking difficulties caused by small targets in remote sensing satellite video tracking, so more than 90% of the targets in this dataset are cars. We compare the performance between our method and other methods on the SatSOT dataset and the SV248S dataset, respectively, and conduct ablation experiments on the SatSOT dataset.

2) Evaluation Metrics: We adopt one-pass evaluation (OPE) and evaluate the performance of all object tracking methods using precision and success metrics. The precision metric measures the center location error (CLE)

$$\mathrm{CLE} = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}. \tag{17}$$

When the Euclidean distance between the center of the predicted bounding box (x_1, y_1) and the ground-truth bounding box (x_2, y_2) is less than a given threshold, the tracking is considered successful. Since most satellite video objects are very small, we set the threshold to five pixels, while the threshold for general video object tracking datasets, such as OTB [8], GOT10k [42], and LaSOT [41], is 20 pixels. The success metric measures the overlap ratio between the predicted bounding box B_p and the ground-truth bounding box B_g. The overlap ratio is defined as follows:

$$S = \frac{B_g \cap B_p}{B_g \cup B_p}. \tag{18}$$

To show the algorithm more comprehensively, we also report the average FPS of all trackers.

3) Implementation Details: We implemented the proposed framework SiamMDM on the open source deep learning library PyTorch and executed it on a 64-bit Ubuntu 20.04 workstation with 12-GB memory GeForce RTX3060 GPU. We adopted AlexNet as the backbone for feature extraction, which was trained on the VISO dataset [65]. Specifically, we randomly select five image pairs from each video sequence in the VISO dataset and feed them into the Siamese network for training. The network was trained end-to-end by stochastic gradient descent for 50 epochs. The size of the initial frame template image and the DT image are 127×127, and the size of the search area image is 255×255. The batch size was set to 8. The learning rate, momentum, and weight decay were set to 0.005, 0.9, and 0.0001, respectively.

In addition to some basic parameters in the Siamese network, there are four preset hyperparameters in our method, which are θ_1, θ_2, and η in the DT update strategy and N in the MM, respectively. After a series of comparative experiments, we finally locked these four parameters at $\theta_1 = 0.75$, $\theta_2 = 2$, $\eta = 0.8$, and $N = 30$.

B. Ablation Study

To verify the effectiveness of each module, we conducted an ablation study on the SatSOT satellite video dataset. The results of the ablation experiments are shown in Table II. "✓" indicates that the module is used in the Siamese network, while "−" indicates that the module is not used in the Siamese network. The first M of the network suffix represents the MRMF strategy, the D in the suffix represents the DT update strategy, and the second M represents the target motion trajectory MM.

The ablation study shows that the MRMF strategy (M), DT update strategy (D), and motion trajectory prediction model (M) in our method all play an essential role in the algorithm.

In addition, to verify that our proposed network can effectively handle the object appearance change problem, we conducted ablation experiments on video sequences containing the ROT attribute and the partial occlusion (POC) attribute, both of which lead to changes in the appearance of the object, respectively. Of these, there are 56 video sequences labeled with the ROT attribute and 34 with the POC attribute. As we can see from Table II, our method shows a good improvement in tracking performance relative to baseline on both videos with the ROT attribute and videos with the POC attribute. Furthermore, in adding the three proposed modules to the

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TABLE I
DESCRIPTIONS OF 11 ATTRIBUTES AND DEFINITIONS IN SATSOT

Attribute	Definition
Background Clutter	The background has similar appearance as the target
Illumination Variation	The illumination of the target region changes significantly
Low Quality	The image is low quality and the target is difficult to be distinguished
Rotation	The target rotates in the video
Partial Occlusion	The target is partially occluded in the video
Full Occlusion	The Target is fully occluded in the video
Tiny Object	At least one ground truth bounding box has less than 25 pixels
Similar Object	There are objects of similar shape or same type around the target
Background Jitter	Background jitter brings by the shaking of satellite camera
Aspect Ratio Change	The ratio of the bounding-box aspect ratio of the first and the current frame is outside the range [0.5, 2]
Deformation	Non-rigid object deformation

TABLE II
ABLATION STUDY OF DIFFERENT COMPONENTS OF SIAMMDM ON THE SATSOT DATASET (MRMF: MULTIPLE RESPONSE MAP FUSION, DT: DYNAMIC TEMPLATE, MM: MOTION MODEL, ROT: ROTATION, AND POC: PARTIAL OCCLUSION)

Method	MRMF	DT	MM	All (105)		ROT (56)		POC (34)		Speed (FPS)
				Prec. (%)	Succ. (%)	Prec. (%)	Succ. (%)	Prec. (%)	Succ. (%)	
SiamFC(Baseline)	-	-	-	49.8	41.3	45.4	33.6	34.3	33.3	**173.5**
SiamFC-M	✓	-	-	55.9↑6.1	42.9↑1.6	53.5↑8.1	37.4↑3.8	38.5↑4.2	33.4↑0.1	140.2
SiamFC-MD	✓	✓	-	58.2↑8.4	45.4↑4.1	58.3↑12.9	40.3↑6.7	40.2↑5.9	35.2↑1.9	111.8
SiamMDM(ours)	✓	✓	✓	**60.5**↑10.7	**47.0**↑5.7	**61.2**↑15.8	**42.2**↑8.6	**41.5**↑7.2	**36.0**↑2.7	110.4

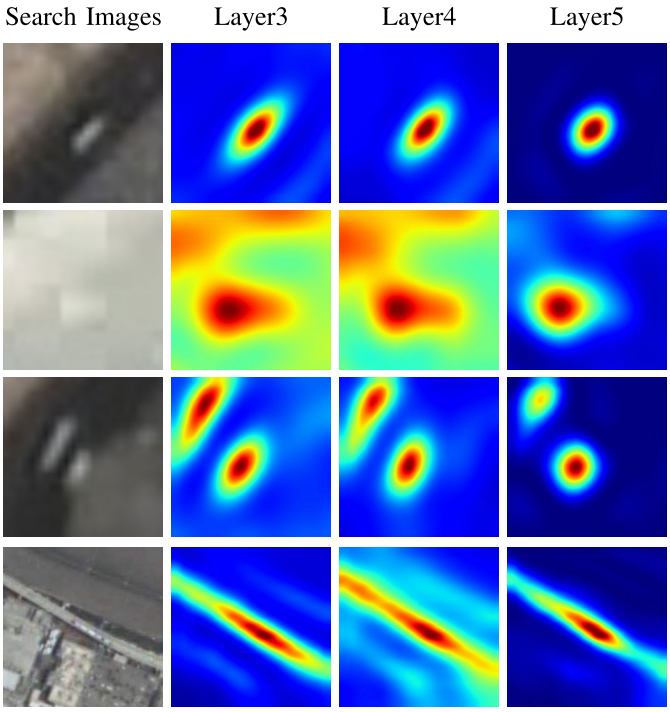


Fig. 6. Visualization of response maps from different layers.

baseline, the tracking performance of the network increases on videos with ROT attribute and videos with POC attribute, demonstrating that all three new proposed modules are able to mitigate the object appearance change problem due to challenging attributes, such as ROT and POC.

1) MRMF Strategy: As mentioned earlier, in the MRMF module, we are fusing the response maps from different locations of the Siamese network to get a fused response map. In addition to fusing all the response maps from the third, fourth, and fifth layers, we can also look at the effect of fusing other layers. The specific meaning of Layer3, Layer4, and Layer5 in Table III refers to the response maps obtained from the third, fourth, and fifth layers of the Siamese network. Table III shows that if we only fuse the response maps from layers 3 and 4, the final result will have a significant point drop relative to the final network result. The main reason is that the network will not have enough semantic information if we do not use the high-level semantic information from layer 5. Also, this will lead to the network not being robust enough in the face of deformation and ROT. Conversely, if we fuse the response maps from layers 4 and 5 or fuse the response maps from layers 3 and 5, the network can utilize the deep semantic information from the deep layers and obtain detailed information from the shallow layers. Hence, the combination of shallow response map and deep response map can achieve better results.

TABLE III
OBJECTIVE RESULTS WITH FUSED RESPONSE MAPS FROM DIFFERENT LOCATIONS (AC: ADAPTIVE COEFFICIENT)

Layer3	Layer4	Layer5	AC	Prec. (%)	Succ. (%)	FPS
✓	✓	-	-	54.1	42.6	110.7
✓	-	✓	-	56.8	44.1	110.7
-	✓	✓	-	56.8	44.0	**111.5**
✓	✓	✓	-	56.2	43.8	110.6
✓	✓	✓	✓	**60.5**	**47.0**	110.4

In order to show the differences between the response maps from the different layers, we visualized the response maps in Fig. 6. In the penultimate row of subplots, the shallow response maps are able to maintain good spatial detail when similar objects appear in the same image, but cannot

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TABLE IV
EFFECT OF DT UPDATE FREQUENCY AND THRESHOLD θ_1 ON THE FINAL RESULT

	Method	K=2	K=5	K=10	K=25	K=50
θ_1 = **0.75**	**Prec. (%)**	58.4	58.5	**60.5**	59.8	59.1
	Succ. (%)	45.7	45.7	**47.0**	46.6	46.4
	FPS	91.6	105.2	110.4	115.6	**120.1**
θ_1 = **0**	**Prec. (%)**	56.5	58.2	60.2	59.5	58.7
	Succ. (%)	43.6	45.4	46.5	46.1	45.8
	FPS	86.1	102.5	109.9	115.1	119.8

TABLE V
FINAL TRACKING RESULTS CORRESPONDING TO DIFFERENT VALUES OF η IN THE DT UPDATE MODULE

Value of η	**0.3**	**0.4**	**0.5**	**0.6**	**0.7**	**0.8**	**0.9**
Prec. (%)	58.1	58.5	60.3	59.1	59.6	**60.5**	59.5
Succ. (%)	44.2	45.5	46.5	45.5	46.3	**47.0**	46.2

TABLE VI
FINAL TRACKING RESULTS CORRESPONDING TO DIFFERENT TRAJECTORY LENGTHS N IN THE MM

Value of N	**10**	**20**	**30**	**50**	**100**	**200**
Prec. (%)	59.6	59.3	**60.5**	60.4	60.1	59.2
Succ. (%)	46.5	46.5	**47.0**	47.0	46.6	46.4

distinguish the target from the similar surrounding objects. The deeper response maps are able to distinguish well between the target and similar distractions, but there is a reduction in the retention of spatial detail. In summary, the shallow response maps maintain more detail, while the deeper response maps have more semantic information and are more discriminative. Therefore, an appropriate combination of deep and shallow response maps can make better use of both the spatial detail and the semantic information in the feature maps.

In addition, we also compare the difference between using adaptive coefficients (ACs) and not using them. The fourth and fifth lines in Table III both fuse the response maps from layers 3–5. The difference is that in line 4, we give the same weight coefficients to each response map. The weight coefficients are 1/3 for each layer of the response map, while the coefficients in front of each response map in line 5 are obtained by network learning.

We can find that if we just fuse the three response maps equally, it is not as good as just combining the deepest fifth layer and the shallower previous layer equally. The possible reason is that the three response maps do not contribute equally to the final result. If three response maps are fused together by learning the weight coefficients through the network, the tracking result will be the best among them.

2) DT Update Strategy: For the DT update module, we conducted an experimental analysis on the frequency of updating the DT and the threshold value for updating the DT. The experimental results are shown in Table IV.

The results show that when updating frequently, e.g., every two frames, the effect of selecting DTs with filtering is better than that of unconditionally updating DTs directly. When the frequency of updating templates decreases, e.g., every 50 frames, the difference between setting the threshold and not is insignificant. At the same time, we find that updating the DT at a lower frequency is better than updating it very frequently. This can be interpreted as that after exceeding a specific update frequency, the faster updating the DT, the more likely the template will be contaminated, leading to a decrease in the final result.

When K = 10 and set the threshold value for updating the template, the best results are achieved.

We also find that the conditional update of the DT is faster than the unconditional update of the DT. The faster the update frequency, the more obvious the tracking speed gap. This is because some unqualified templates can be eliminated through the coefficient θ_1.

On top of that, we analyzed the effect of different η values on the final tracking results in detail by comparative experiments, and the results are shown in Table V. The parameter η is used to balance the prediction results of the initial template branch with those of the DT branch, and it takes values between [0, 1]. The closer the η is to 1, the more the network weighs the prediction results of the initial template branch, and vice versa, the more the network weighs the prediction results of the DT branch.

Through Table V, we find that the tracker achieves the best tracking results when $\eta = 0.8$, where the precision rate of tracking is 60.5% and the success rate of tracking is 47.0%. It indicates that when combining the prediction results of the initial template branch and the prediction results of the DT, the prediction results of the initial template branch should be dominant, and the prediction results of the DT branch should be supplementary. When $\eta = 0.3$, the precision rate of tracking is 58.1%, which is 2.4% lower than the best result; the success rate of tracking is 44.2%, which is 2.8% lower than the best result. It indicates that the initial template contains a lot of useful information, and overemphasizing the role of the DT while ignoring the information contained in the initial template can cause a significant decrease in tracking effectiveness. When $\eta = 0.9$, the precision rate of tracking is 59.5%, which is 1.0% lower than the best result, and the success rate of tracking is 46.2%, which is 0.8% lower than the best result. This indicates that the role of DTs is also indispensable and it needs to take a place in the final decision.

In summary, the DT branch is indispensable. While the Siamese network mainly refers to the prediction results of the initial template branch, it also needs to appropriately refer to the prediction results of the DT branch.

3) Motion Model: We conducted an ablation experiment on the length of N in the motion trajectory prediction model module. N indicates how many previous frames are used to predict the target's location in the next frame when using this MM. As demonstrated in Table VI, as the length of N increases, the precision rate and success rate show a general increase and then decrease.

The reason for this is that, if we only use few previous frames to predict target position in current frame, the prediction results of the MM are likely to be influenced by the

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TABLE VII

OVERALL PERFORMANCE ON SATSOT AND SV248S. THE BEST THREE PERFORMANCES ARE, RESPECTIVELY, HIGHLIGHTED WITH **RED**, **GREEN**, AND **BLUE** COLORS (FOR FEATURES, GFS: GRAY FEATURES, HOG: HISTOGRAM OF ORIENTED GRADIENT, CNS: COLOR NAMES, CONVFEAT/CF: CONVOLUTIONAL FEATURE, AND TF: TRANSFORMER)

Trackers	Methods	Source	Features	SatSOT		SV248S		Speed (FPS)
				Prec. (%)	Succ. (%)	Prec. (%)	Succ. (%)	
MOSSE	CF-based	CVPR2010	GF	24.2	26.9	16.6	7.8	156.3
CSK	CF-based	ECCV2012	GF	23.7	24.7	8.9	5.0	131.6
KCF	CF-based	TPAMI2014	HOG	52.1	39.3	43.6	19.6	**187.0**
Staple	CF-based	CVPR2016	HOG	46.2	38.2	34.3	14.7	38.0
ECO	CF-based	ICCV2017	HOG+CN+CF	54.9	38.7	67.7	40.6	46.6
BACF	CF-based	ICCV2017	HOG	55.0	**44.2**	**69.6**	25.8	18.6
STRCF	CF-based	CVPR2018	HOG	53.0	41.0	68.1	22.9	15.4
SiamFC	DL-based	ECCVW2016	ConvFeat	49.8	41.3	63.4	39.4	173.5
DCFNet	DL-based	arXiv2017	ConvFeat	54.1	42.6	59.0	23.6	55.7
SiamRPN	DL-based	CVPR2018	ConvFeat	50.0	38.5	34.0	14.5	**320.5**
ATOM	DL-based	CVPR2019	ConvFeat	52.8	42.4	62.6	36.3	48.3
SiamRPN++	DL-based	CVPR2019	ConvFeat	54.0	40.0	67.3	**42.4**	45.1
SiamRPN++_lt	DL-based	CVPR2019	ConvFeat	52.8	38.7	55.1	31.9	41.9
SiamMask	DL-based	CVPR2019	ConvFeat	**55.2**	39.8	56.5	22.1	83.6
DiMP18	DL-based	ICCV2019	ConvFeat	51.8	42.0	59.6	36.0	58.8
DiMP50	DL-based	ICCV2019	ConvFeat	52.8	42.6	60.0	36.0	37.6
SiamFC++	DL-based	AAAI2020	ConvFeat	44.8	34.5	40.3	21.3	**216.7**
PrDiMP18	DL-based	CVPR2020	ConvFeat	46.5	40.2	56.8	35.9	39.7
TransT	DL-based	CVPR2021	CF+TF	49.6	38.8	49.2	27.3	33.3
ToMP50	DL-based	CVPR2022	CF+TF	49.3	38.9	38.8	16.5	20.3
ToMP101	DL-based	CVPR2022	CF+TF	46.5	36.9	37.2	16.0	15.7
TCTrack	DL-based	CVPR2022	CF+TF	45.7	32.9	32.5	13.0	125.6
SiamMDM	DL-based	Ours	ConvFeat	**60.5**	**47.0**	**72.0**	**45.3**	110.4
SiamMDM50	DL-based	Ours	ConvFeat	**61.1**	**47.5**	**72.5**	**45.6**	61.5



(a) (b)

Fig. 7. (a) Precision score of different attributes on the SatSOT dataset. (b) Success score of different attributes on the SatSOT dataset.

wild points in the previous target locations. However, if too many previous frames are applied to predict the target position in current frame, it will also lead to precision degradation, because the targets in satellite video generally only conform to uniform linear motion in a short time. The satellite videos in the dataset are taken at 10 FPS or 25 FPS with about a 30-s duration, which means many objects are not simply doing uniform linear motion. So, the too long length of N will lead to a decrease in tracking accuracy. However, even if the appropriate parameters are not selected, we can still find through the data that the final results obtained by using the MM are still better than those obtained by not using the MM. Also, when N equals 30, the network achieves the most competitive result.

C. Comparison With State of the Art

1) Evaluated Trackers: To demonstrate the superiority of our method, an evaluation of 22 representative trackers is conducted on the SatSOT dataset and the SV248S dataset. We compare SiamMDM with seven representative traditional methods (MOSSE [12], CSK [13], Staple [66], KCF [15], ECO [18], BACF [67], and STRCF [68]), 11 convolutional neural network methods (SiamFC [26], SiamRPN [9], SiamFC++ [69], ATOM [70], SiamRPN++ [34], SiamMask [32], SiamRPN++_lt [34], DCFNet [71], DiMP18 [72], DiMP50 [72], and PrDiMP18 [73]), and four TF methods (TransT [36], TCTrack [39], ToMP50 [74], and ToMP101 [74]).

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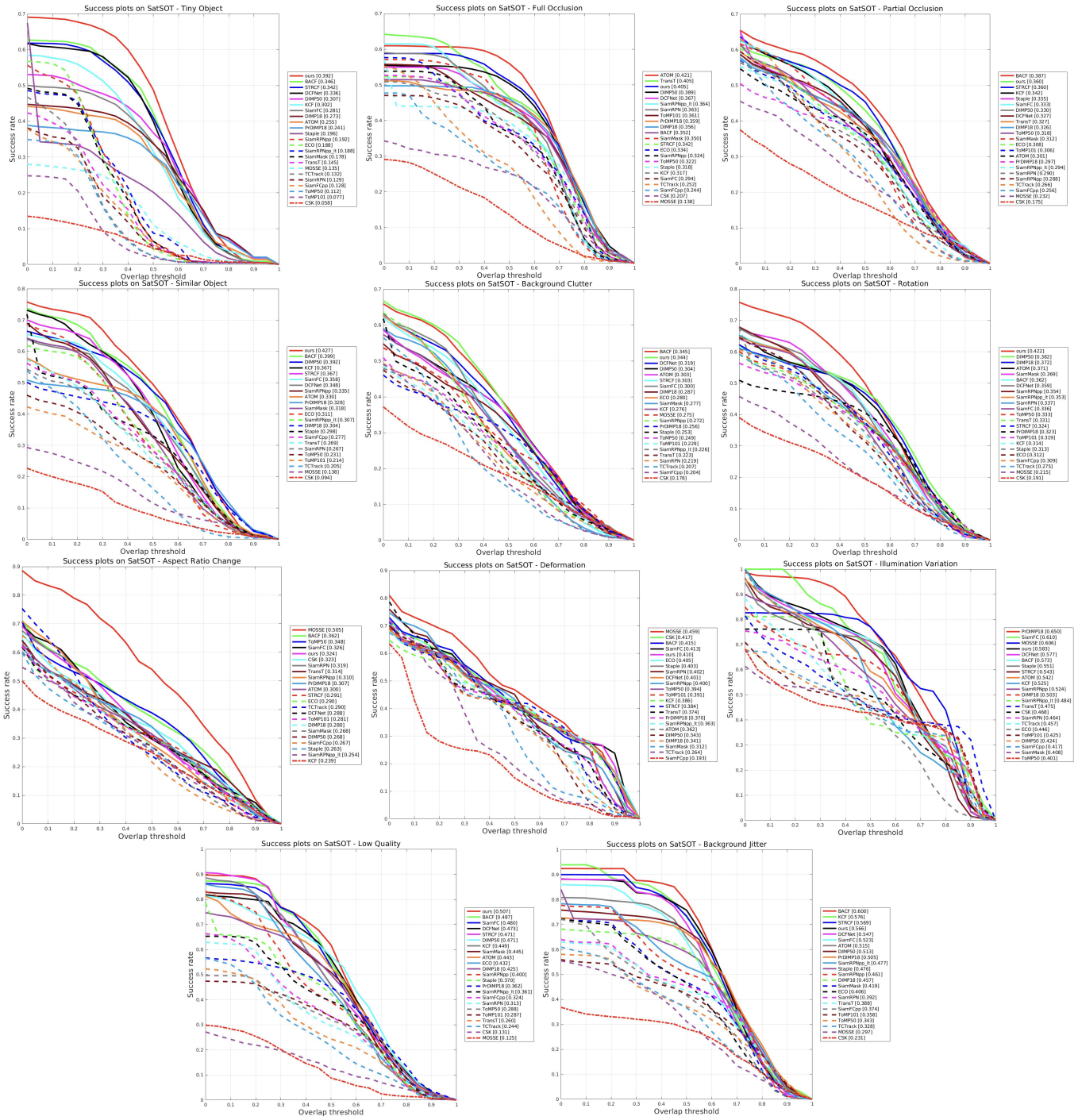


Fig. 8. Performance of trackers on each attribute using success.

Among them, all deep learning-based methods are tested with their respective official provided models to obtain the best tracking results. Although we retrained all these deep learning-based methods on the tracking dataset related to satellite images, almost all of them tested with worse results than the officially provided models. The literature [64] comes to the same conclusion. The possible reason is that these deep learning-based methods are designed for general-purpose video target tracking and do not take into account factors, such as smaller target size in satellite video, so retraining the general-purpose tracking network with the satellite video tracking dataset does not yield better tracking results. Therefore, we ended up using the officially provided models for testing, as what was did in the literature [64], to get the best performance.

2) Overall Results on SatSOT and SV248S: Table VII shows the precision rate, success rate, and speed of all algorithms on the SatSOT and SV248S satellite video datasets. SiamMDM denotes the proposed method using AlexNet as the backbone, while SiamMDM50 denotes the proposed method using the modified ResNet-50 as the same in [34] as the backbone. Our proposed SiamMDM50 yields state-of-the-art tracking

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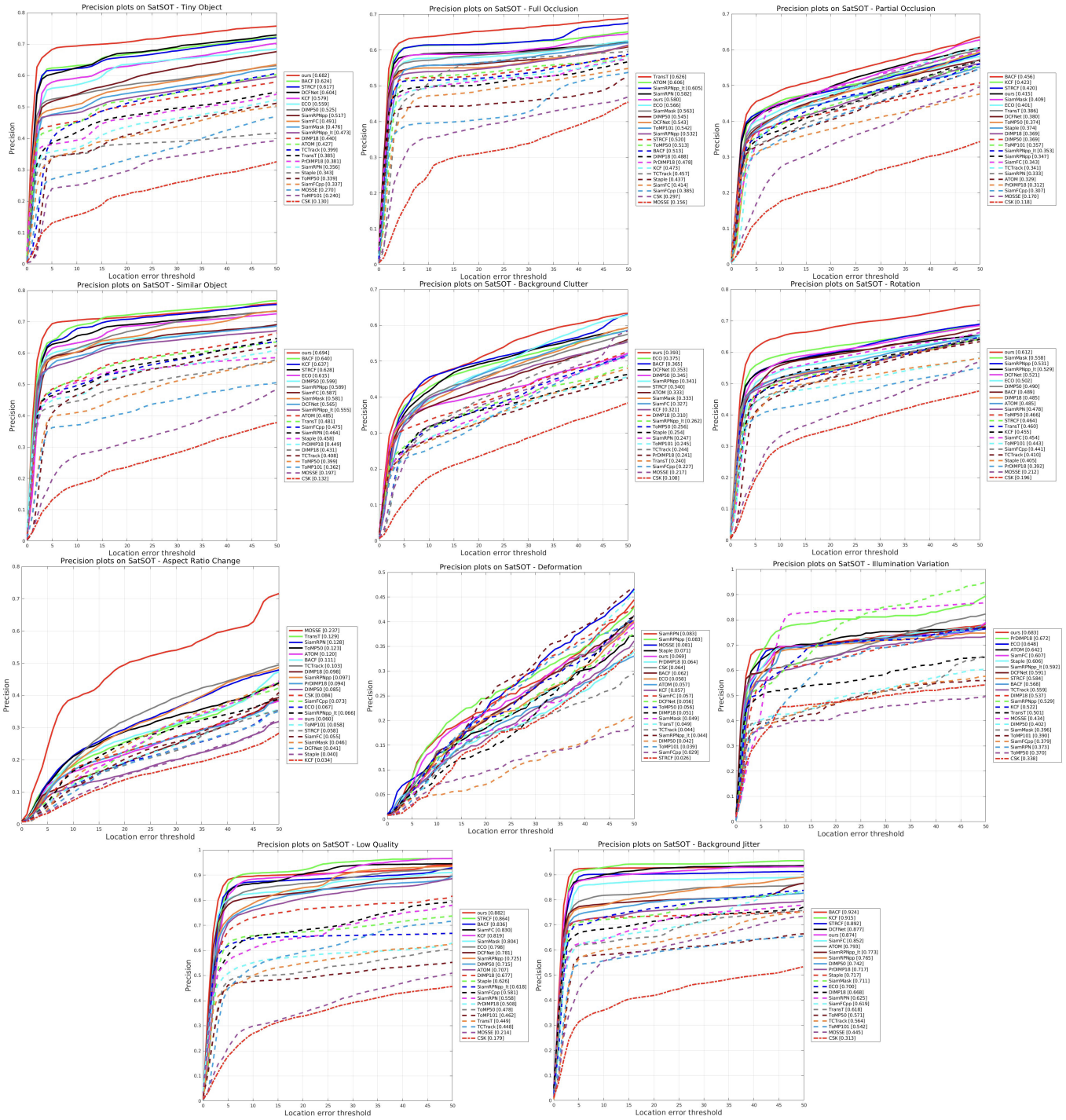


Fig. 9. Performance of trackers on each attribute using precision.

performance in both precision and success rate dimensions, while our lightweight version SiamMDM strikes a balance between speed and performance. It is worth mentioning that since the targets in the satellite video are small and the backbone can only extract limited features, the use of a deeper backbone has a relatively small impact on the improvement of the final tracking results, unlike in the field of general-purpose target tracking. SiamRPN achieved the fastest tracking speed with an excellent performance of 320.5 FPS. The traditional method BACF has achieved competitive results with precision score 55.0% and success score 44.2% in the SatSOT dataset, but its tracking speed is only 18.6 FPS and fails to reach the real-time line. In recent years, with the rapid development of TF, many algorithms apply TF to general-purpose target tracking. We selected four TF-based general-purpose target tracking methods to be included in the evaluation. We were surprised to find that due to the differences between general-purpose target tracking and satellite video target tracking, some algorithms that work well in general-purpose target tracking perform poorly in satellite video target tracking.

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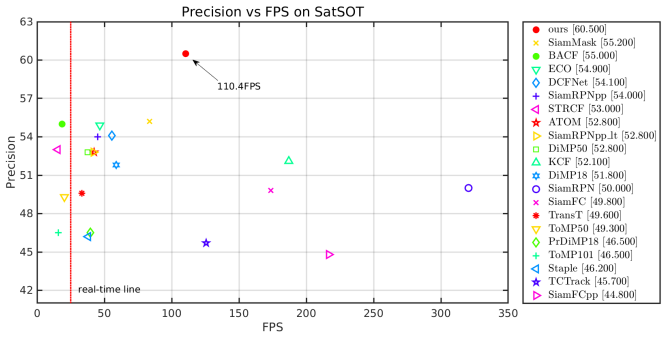


Fig. 10. Comparison of the performance and speed of state-of-the-art tracking methods on SatSOT. We visualize the precision rate with respect to the FPS.

3) Attribute-Based Results on the SatSOT Dataset: For the precision metric, background clutter, aspect ratio change, and deformation are the most challenging tasks among 11 attributes. All trackers get scores no more than 0.4 in these attributes. For clarity, Fig. 7 shows the radar plots for top six trackers on precision and success, respectively. For the precision metric, the proposed SiamMDM method ranks highest in six (ROT, low quality, illumination variation, background clutter, similar object, and tiny object) out of 11 attributes and achieves the first in overall precision scores. For the success metric, the proposed SiamMDM method rank highest in five (ROT, low quality, illumination variation, similar object, and tiny object) out of 11 attributes and achieves the first in overall success scores. To analyze the performance differences of the trackers on each attribute challenge, we compared our algorithm with 11 typical trackers. We plotted the precision score and success score of the trackers on each attribute challenge, respectively, as shown in Figs. 8 and 9. We find that, for the success metric, tiny object, background clutter, and POC are the most challenging tasks among 11 attributes.

In satellite video datasets, the number of small objects is much larger than that of large targets, making tiny object an important challenging attribute. Because of the MRMF strategy, SiamMDM can use multiple response maps to locate small objects more accurately, thus achieving 68.2% in the precision metric and 39.2% in the success metric, both achieving the best results. More importantly, in addition to the initial frame template, the network can obtain a lot of information that is more relevant to the target in the current search frame through DT branch. Therefore, SiamMDM has a better performance in the three challenging attributes of ROT, illumination variation, and background clutter compared with other methods. In addition, with the help of the MM, although similar objects may score close to the tracked target in the feature map, the network can determine where the target is more likely to appear in the current frame from the target's historical position, which helps the network to correctly distinguish between real targets and other similar objects, and also helps to predict objects when the image quality is low and the target is difficult to distinguish from the background. Therefore, SiamMDM achieves the best results in both similar object and low quality.

Nevertheless, our method does not achieve the best in all attributes. In terms of two evaluation metrics of precision and success, our method still has shortcomings in the three challenging attributes of background jitter, aspect ratio change, and POC.

On the attribute of background jitter, SiamMDM scores 87.4% for the precision metric and 56.6% for the success metric. In the SiamMDM method, an essential prerequisite for the excellent performance of the MM auxiliary branch is that the background of the target is relatively stationary. When the background inevitably shakes due to camera or lighting vibration or shaking, the current position of the target predicted by the MM based on the target's historical location will have a significant deviation from the actual position of the target, resulting in a deviation between the final predicted result and the actual position of the target after its correction. This causes SiamMDM to underperform on the challenging attribute of background jitter.

For the attribute aspect ratio change, SiamMDM scores 6.0% for the precision metric and 32.4% for the success metric. When the aspect ratio of a target changes dramatically in a short period of time, the similarity value between the target in the search area and the target in the template area decreases abruptly, which may lead the network to track other targets incorrectly. Since no specific strategy was developed for aspect ratio change, the results of the SiamMDM method for this challenging attribute are relatively average.

Regarding POC, SiamMDM scored 41.5% for the precision metric and 36.0% for the success metric. Almost all deep learning-based methods encountered difficulties with this challenging attribute. One of the possible reasons is that the network does not get enough training data to learn how to handle occlusions. The POC attribute in satellite images is very complex; it could be caused by multiple factors, such as buildings, trees, clouds, and so on, and the Siamese networks have difficulty learning how to handle the occlusion patterns. In addition, most of the Siamese network methods nowadays are learning the similarity between the targets in the search region and the targets in the template region. When random occlusions appear, the targets in the search region may lose most of their information, which makes it difficult for the network to match them with the targets in the template region. Even so, it is worth mentioning that our proposed SiamMDM method is the best-performing deep learning-based method on the challenging attribute of POC, second only to the best-performing correlation filter-based method BACF on this challenging attribute. This is mainly aided by the DT branch and the MM branch.

In the future, we will make further improvements to address these issues.

4) Run-Time Analysis: In order to show the relationship between the speed and the performance of each tracker, we draw a graph of the speed and performance of all target trackers, as shown in Fig. 10.

The red line in Fig. 10 represents the real-time line, which is an essential basis for judging whether the algorithm can run in real time. TF-based deep learning methods are generally slow due to their complex structural characteristics. Since

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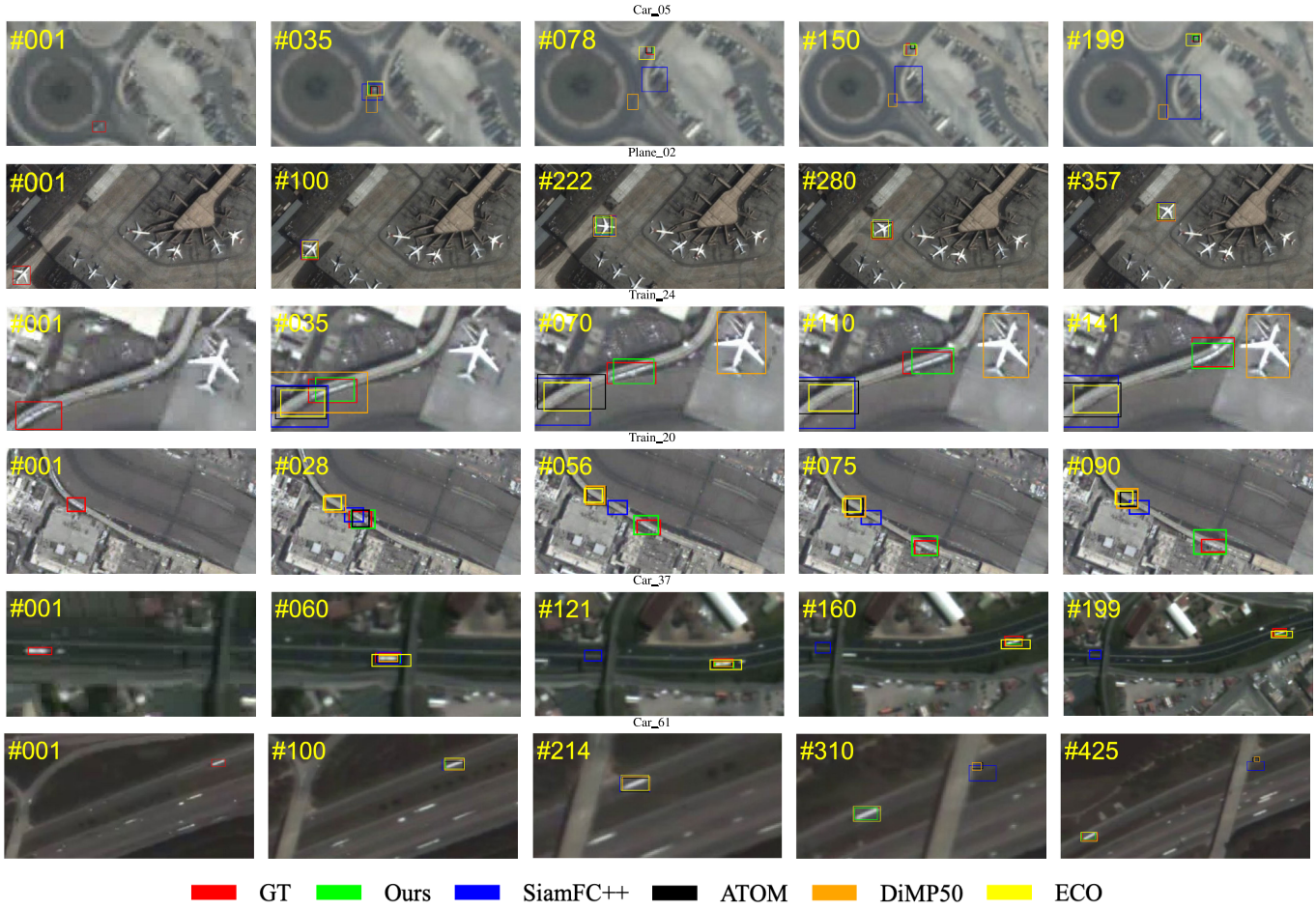


Fig. 11. Qualitative comparison of state-of-the-art visual trackers on the Car_05, Plane_02, Train_24, Train_20, Car_37, and Car_61 video sequences from the SatSOT dataset. The frame number are shown on each frame.

the objects of interest in satellite videos are generally very small, the network cannot adequately extract the features of the objects from the pictures, so the performance of the TF is suppressed. As a result, TF-based deep learning methods are temporarily unable to shine in the field of satellite video tracking. Our proposed SiamMDM method achieves the highest precision based on the mainstream Siamese network structure. It achieves a speed of 110 FPS, far exceeding the 24 FPS required for real-time speed, enabling our algorithm to run in real time while giving a competitive tracking effect.

5) Visual Analysis: In order to qualitatively evaluate the performance differences of different trackers when tracking satellite video targets, five representative trackers are selected for comparison with the ground truth, including SiamFC++, ATOM, DiMP50, ECO, and our proposed method SiamMDM. Six video sequences with different challenging properties from the SatSOT dataset are selected for visual analysis, including Car_05, Plane_02, Train_24, Train_20, Car_37, and Car_61. The visualization results are shown in Fig. 11. The video sequence Car_05 includes three challenging attributes: background occlusion, ROT, and low quality. Due to the low quality of the satellite video images and the lack of significant contrast between the target vehicle's body color and the road, the target vehicle appears to "disappear" in some sequences in the middle of the video. The SiamFC++ algorithm and the DiMP50 algorithm each shifted the tracking to the surrounding roads and roadside objects with a similar color to the target vehicle after the target disappeared. Also, since the target moves out of the tracker's search area after a period of time, they can only fail to recapture the reappearing target. Our algorithm succeeds in capturing the target again when it reappears. The reason is that the score of Siamese network tracking decreases when the target disappears. SiamMDM transfers the weight of the final tracking result to the target trajectory prediction model, which predicts the theoretical motion trajectory after the target disappears based on the motion trajectory when the target has not yet disappeared. So, when the target reappears, the target happens to be near the search area of the tracker, and our tracker can track the target again. The video sequence plane_02 includes two challenge attributes: brightness change and ROT. As shown in the image, the size of each tracker prediction frame varies when the plane rotates in the middle of the video sequence. When the plane rotates again to the same angle as the initial phase of the video, the size of each tracker prediction frame converges to the size of the ground truth. This is because most of the algorithms selected for our visualization analysis use only the first frame as the template frame. When the angle of the tracked object changes, the tracker retains only the target's middle body part and ignores the target's edges. Although our algorithm incorporates a DT strategy, it cannot

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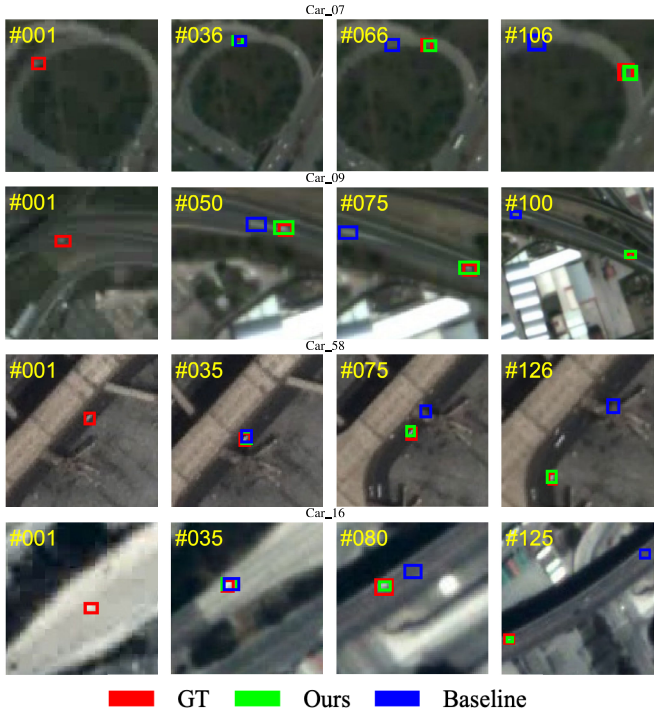


Fig. 12. Comparison of SiamFC (baseline) and SiamMDM (ours) with ground truth in four satellite sequences.

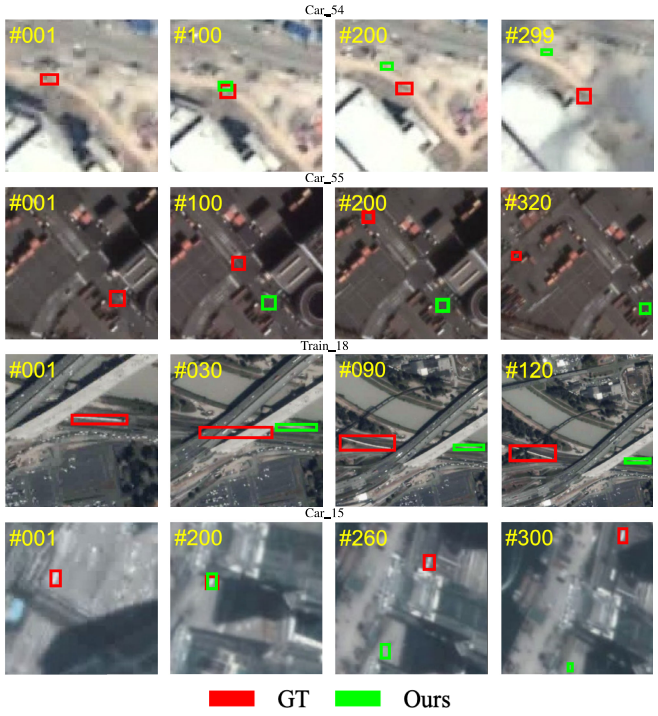


Fig. 13. Failure cases of SiamMDM network in four video sequences.

avoid such a problem because of the high weight of the first frame. In the video sequence Car_37, the biggest challenge for tracking is the POC problem. From the picture, we can find that when passing through the occlusion of the bridge road, SiamFC++ loses the tracked target due to the disappearance of part of the tracked object, and the prediction box always stays at the place before the target enters the bridge road. But, fortunately, since there are no other objects with similar appearance, shape, and size within the search area, other trackers turn the tracking object to the part where the target has drilled out of the bridge road, so other trackers can still track the target normally. In the video sequence Car_61, the biggest challenge for tracking is the total occlusion problem. Unlike the video sequence Car_37, the target vehicle is completely occluded when passing under the bridge road without any part exposed. Before 214 frames, all trackers could accurately track the target vehicle. But, when the target vehicle reappears after being completely occluded by the road bridge, only our method and ECO can track the target vehicle. Similar to the video sequence Car_05, our method uses the motion trajectory prediction model to assist the Siamese network in predicting the theoretical position when the target vehicle disappears.

To demonstrate the effectiveness of our proposed method on the challenge attributes of tiny object, occlusion, and background noise, we selected four video sequences containing these challenge attributes and compared SiamFC (baseline) and SiamMDM (ours) with ground truth in these sequences. The comparison is visualized in Fig. 12. Among them, the critical challenge attribute of Car_07 and Car_09 is tiny object, and the key challenge attribute of the Car_58 video sequence is occlusion, while the critical challenge attribute of the Car_16 video sequence is background noise. The baseline invariably loses the target or follows the wrong target in all four video sequences. Our proposed method can effectively deal with the challenges of small targets, occlusion, and background noise, maintaining the tightness of the bounding box while tracking consistently and correctly.

6) Failure Cases and Model Extensions: Although SiamMDM has achieved good performance in tracking accuracy and speed, there is still room for improvement. We carefully studied the limitations of the SiamMDM network and elaborated on three possible reasons that may lead to tracking failures. In the following paragraphs, we present typical tracking failure cases and discuss possible solutions for each scenario.

a) Absence of effective feature information: As shown in the video sequences Car_54 and Car_55 in Fig. 13, due to the initial frame of video sequence being blurry and the target size being very small, it can be observed that the effective information about the target provided in the initial frame is very limited. There is no significant difference between the target itself and the surrounding background. In this case, the tracker quickly experiences tracking drift. To alleviate the tracking failure caused by the lack of effective information, it is possible to consider preprocessing the image before inputting it into the tracker. For instance, using super-resolution methods to increase the resolution of the original target and enhance its discriminative features [58]. In addition, it is also possible to consider improving the matching method [75] of the Siamese network by using better matching operators to increase the tracker's ability to distinguish the target from the surrounding background.

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b) Absence of movement information: In some video sequences, the target is occluded from the very beginning, which makes it difficult for the network to effectively utilize the target's movement information in the previous frames to predict its position in the subsequent frames. As shown in video sequence Train_18 in Fig. 13, the train is occluded by the road after the first frame and does not emerge until the 90th frame. In this case, the Siamese network may exhibit displacement in predicting the target's center position in the early frames of the video, rendering the MM unable to function properly. Consequently, the tracker may shift the predicted tracking result to the position of similar objects surrounding the target. To address this situation, more motion information can be incorporated into the Siamese network's prediction, such as through optical flow [76] or frame difference [65] methods to better perceive motion information in the images.

c) Multiple challenging factors in combination: In some cases, remote sensing satellite video sequences are faced not only with a single primary challenge but also with various challenging factors, such as tiny object size, background jitter, and long-term full occlusion. For instance, as shown in video sequence Car_15 in Fig. 13, SiamMDM successfully completed the continuous tracking task of the target in the first 200 frames of the video sequence. However, the target was completely occluded between frames 200 and 260, during which time the background underwent significant displacement. This implies that the tracking result of the Siamese network and the prediction result of the MM were both unreliable, resulting in inevitable tracking failure. To alleviate this issue, a local-to-global search strategy [77] can be attempted. When the target is occluded for an extended period, a global search strategy can be used to increase the search area.

V. Conclusion

In this article, we propose an object tracking network based on multiple response map fusion (MRMF) and spatiotemporal constraints, specifically for tracking moving objects in satellite videos. The network obtains fine-grained spatial details and accurate spatial localization by fusing shallow and deep feature response maps. Moreover, we propose a dynamic template (DT) strategy to solve the problem of template changes caused by partial occlusion (POC) and motion blur, which are often seen in satellite videos. We also propose a score-guided motion model (MM) to deal with the tracking drift problem. We conduct extensive experiments on SatSOT and SV248S, two real-world satellite video object tracking datasets. Experimental results show that, the three modules proposed in this article greatly increase the accuracy and robustness of satellite video target tracking. When compared with other state-of-the-art trackers, our proposed SiamMDM tracker achieves best performance and can run at a high speed of 110 FPS. In the future, we will consider integrating super-resolution methods and better target trajectory prediction methods into the tracking process to achieve more competitive tracking performance.

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